

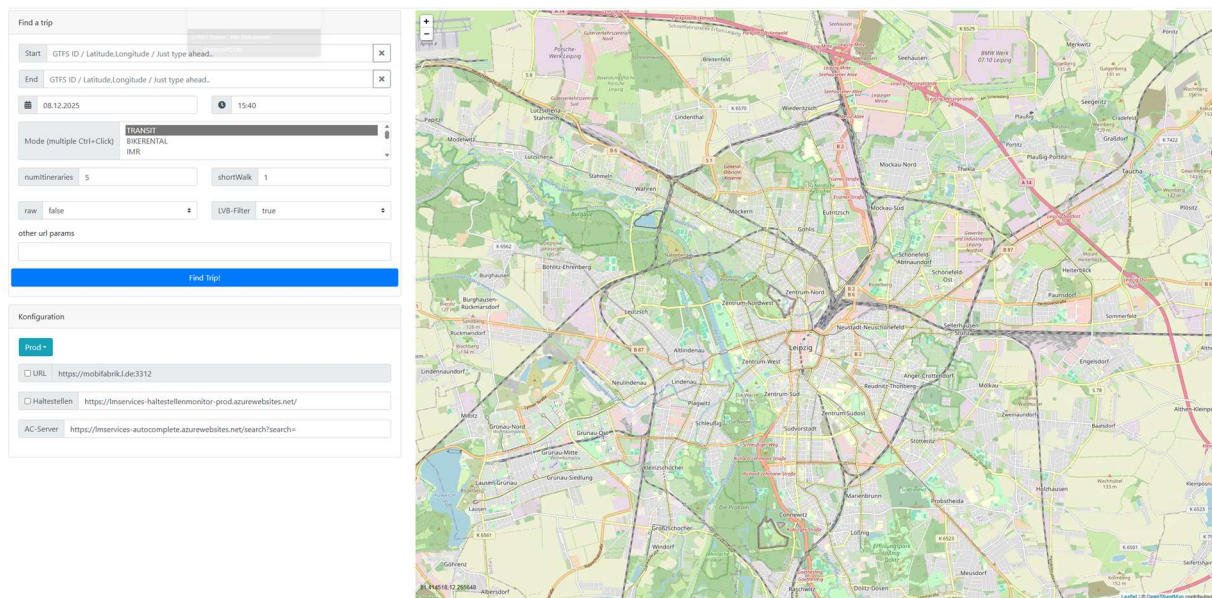
OTP Workbench – A Tool to Test and Compare the LVB Routing APIs

1 Introduction

Leipziger Verkehrsbetriebe (LVB) is the public transport provider of the city of Leipzig. A core part of its service portfolio is the provision of a digital journey planner. LVB offers this functionality via the LeipzigMOVE mobile application and via a web-based journey planner on its website.

The routing used in LeipzigMOVE is based on OpenTripPlanner (OTP) and developed by the LVB routing team. OpenTripPlanner is an open-source routing engine that computes itineraries combining public transport, walking, cycling, and car segments.

Routing developers at LVB frequently need to test routing APIs and compare responses across multiple environments (PROD, STAGE, DEV), locally running instances, and third-party routing services. Currently, this is done using the existing “OTP Client”, which provides limited testing functionality.



The goal of this project is to design and implement **OTP Client v2.0**, a more powerful tool that enables systematic testing and direct comparison of routing APIs.

2 Scope and Language

The application shall be implemented in English and German.

The primary target audience shall be routing developers and technical staff working with routing APIs.

3 General Use Cases

3.1 Supported Scenarios

As a routing developer, the user shall be able to:

- Test a single routing API – *Routing* use case
- Compare multiple routing APIs – *Routing Comparison* use case
- Access all routing parameters in a central parameter area

The following future extensions shall be considered in the GUI design, but are explicitly out of scope for this project:

- Comparison of Autocomplete APIs
- Comparison of Stopmonitor APIs
- Comparison of NearBySearch APIs

There shall be GUI elements to access those future test cases which shall not be functional yet.

3.2 Application Layout

The application shall be structured as follows:

- A Parameter Area on the left side
- An Evaluation Area on the right side
- A navigation area to switch between different use cases

The Evaluation Area shall provide a tab-based navigation for different testing scenarios.

Initially, the *Routing* use case shall be selected and displayed.

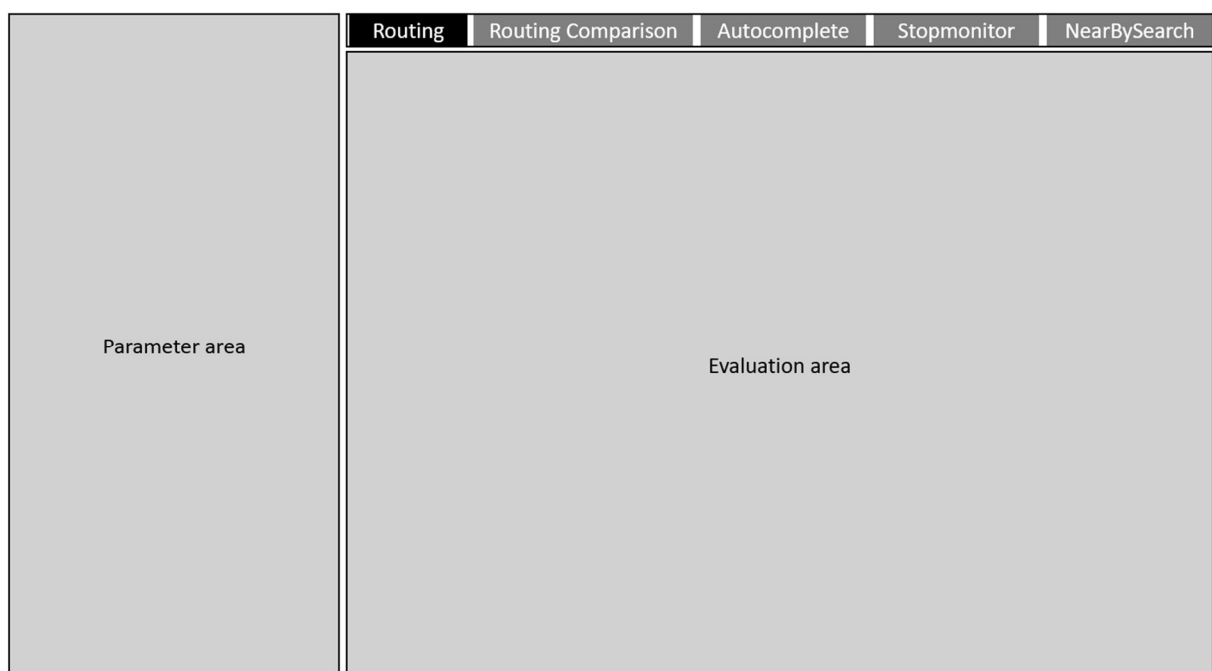


Figure 1: General Layout

4 Parameter Area Requirements

4.1 Environment Selection

The user shall be able to:

- Select routing environments from a predefined list:
 - PROD
 - STAGE
 - DEV
- Add custom environments
- Select up to three environments for comparison
- Mix predefined and custom environments

When the use case *Routing* is selected, the selection of more than one environment shall not be enabled.

Initially, PROD shall be preselected.

4.2 Journey Definition

The user shall be able to define a journey with the following mandatory inputs:

- Start location
- Destination
- Date and time

Start and destination shall be definable by:

- Using the LVB autocomplete
- Entering stop IDs
- Entering geographic coordinates

The system shall provide the following convenience features:

- Selection from the last 10 previously used start/destination inputs
- One-click clearing of start and destination fields
- One-click switching of start and destination

4.3 Time and Routing Options

The user shall be able to:

- Select whether the routing request is based on *depart at* or *arrive by*
- Select one or more travel modes from a predefined list – see apidocs for the list
- Ensure that at least one travel mode is selected

Initially, *depart at* and travel mode TRANSIT shall be preselected.

The user shall be able to optionally set additional routing parameters, including but not limited to:

- shortWalk
- lessTransfers

The user shall be able to enter additional custom parameters in a free-text input field.

4.4 Request Handling and History

The system shall:

- Allow the user to start a routing request based on the entered parameters
- Validate mandatory inputs before sending a request
- Display an appropriate error message if mandatory fields are missing

The system shall maintain a history of the last 20 routing requests. It is sufficient to save the requests in the browser.

The user shall be able to:

- Load a historical request from the list of 20 maintained requests
- Overwrite all current inputs with the loaded request without additional warning

The system shall allow the user to generate a shareable link containing all routing parameters.

4.5 Parameter Area Layout Control

The user shall be able to:

- Collapse the Parameter Area to increase space for the Evaluation Area
- Expand the Parameter Area to modify parameters

5 Routing

5.1 Evaluation Area Overview

For the use case *Routing*, the Evaluation Area shall consist of:

- A Map Area
- A Routing Results Area

5.2 Map Interaction (Applicable to All Routing Scenarios)

The system shall provide the following map interactions:

- Display start and destination points
- Transfer coordinates to the start field on left-click if the start is empty
- Transfer coordinates to the destination field on left-click if the start is filled and the destination is empty
- Display a popup with coordinates if both start and destination are filled
- Allow copying of coordinates
- Display coordinates under the mouse cursor

Before any routing request is executed, the map shall occupy the entire Evaluation Area.

Initially, the map shall be centered on Leipzig main station.

5.3 Layout Options

After a routing request, the system shall display both the map and routing results.

The user shall be able to switch between the following layouts:

- Map above routing results
- Map next to routing results

Initially, the *map above routing results* layout shall be selected.

The user shall be able to resize both areas.

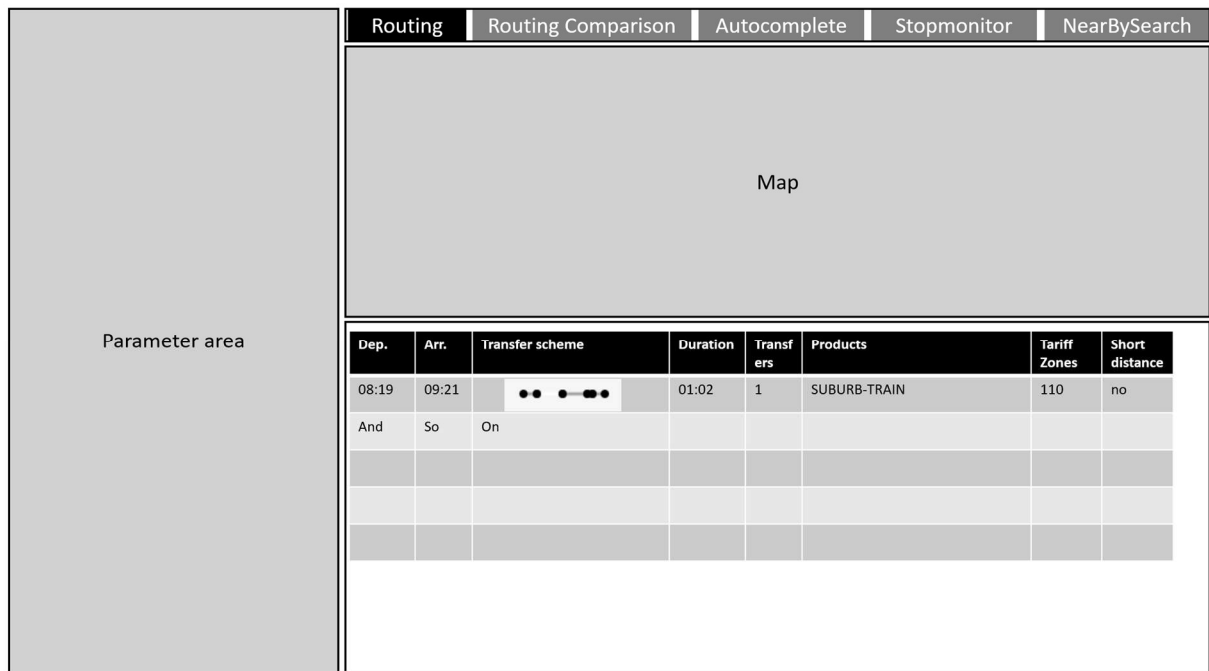


Figure 2: Routing - Map above routing results

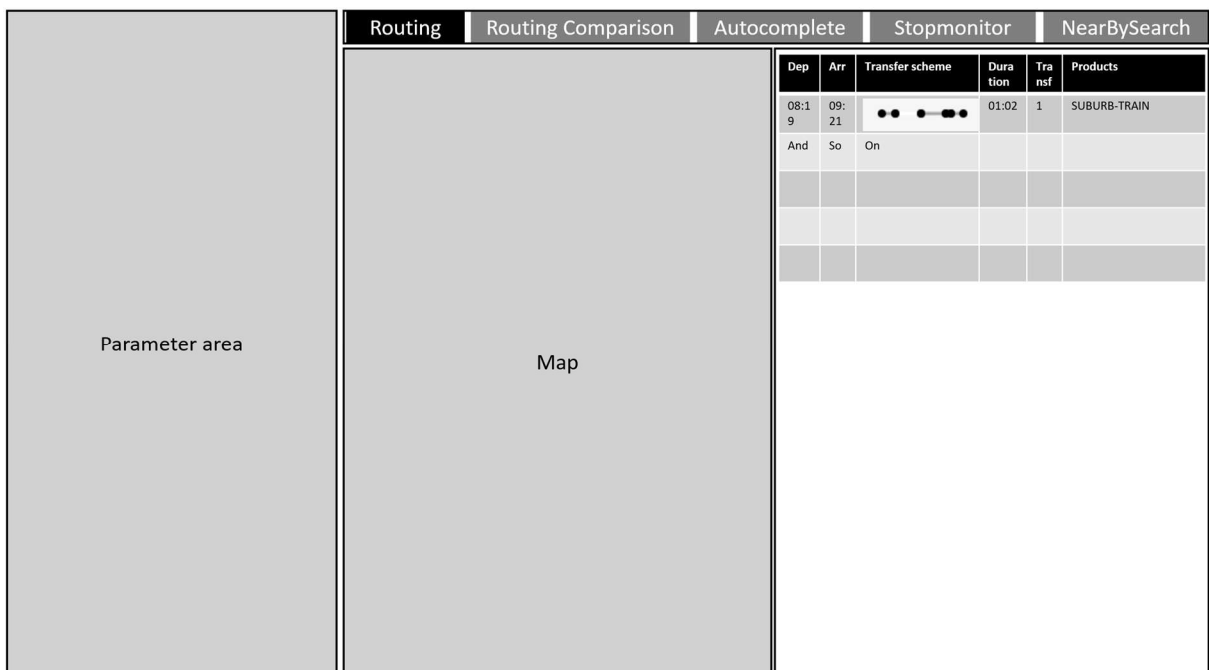


Figure 3: Routing - Map next to routing results

5.4 Routing Results Overview

The system shall display an overview of all received itineraries, including:

- Planned and realtime departure time
- Planned and realtime arrival time
- Transfer scheme
- Total duration

- Number of transfers
- Used transport products

When results are displayed below the map, the system shall additionally display in the overview:

- Tariff zone information for the trip
- Short-distance trip indicator

5.5 Itinerary Detail View

For each itinerary, the system shall provide a detailed view including:

- All legs with start and destination stops (including stop IDs)
- Intermediate stops (expandable and collapsible) incl. stop IDs
- Planned and realtime times per leg
- Duration and products per leg
- Tariff zones per stop
- Additional information (walking distance, trip ID, warnings, info messages)

The route shape shall be displayed on the map with distinct coloring per leg.

Hovering a leg shall highlight the corresponding shape on the map.

5.6 Advanced Itinerary Information

The system shall allow the user to:

- View the complete course of a trip for a leg
- Highlight only the portion relevant to the itinerary
- Access the formatted JSON response of an itinerary
- Copy the JSON response easily
- Load earlier or later itineraries

6 Routing Comparison

(Routing Comparison sections follow the same structure and behavior as Routing unless explicitly stated otherwise.)

6.1 Comparison Layout Modes

The system shall support the following comparison layouts:

- Side-by-side horizontal
- Side-by-side vertical
- Horizontal overview

Initially, *side-by-side horizontal* shall be selected.

6.2 Side-by-Side Horizontal

- One column shall be displayed per selected environment
- A vertical timeline with synchronized scrolling shall be displayed
- Itineraries shall be ordered by realtime departure time
- Transfer schemes shall visualize relative duration

Hovering shall display detailed information and update the map.

Clicking shall open the itinerary detail view.

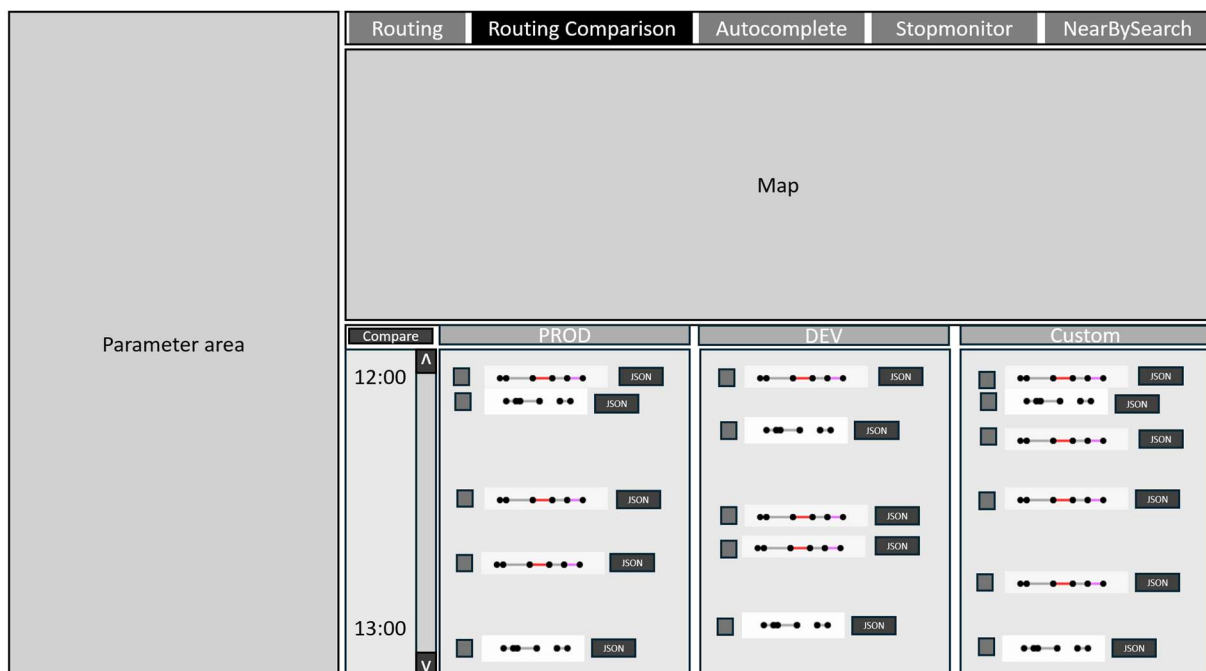


Figure 4: Routing Comparison - Side-by-side horizontal

6.3 Side-by-Side Vertical

- One column shall be displayed per environment
- Transfer schemes shall be displayed vertically on the timeline
- Length and position shall reflect realtime duration and timing

All interaction behavior shall match the horizontal layout.

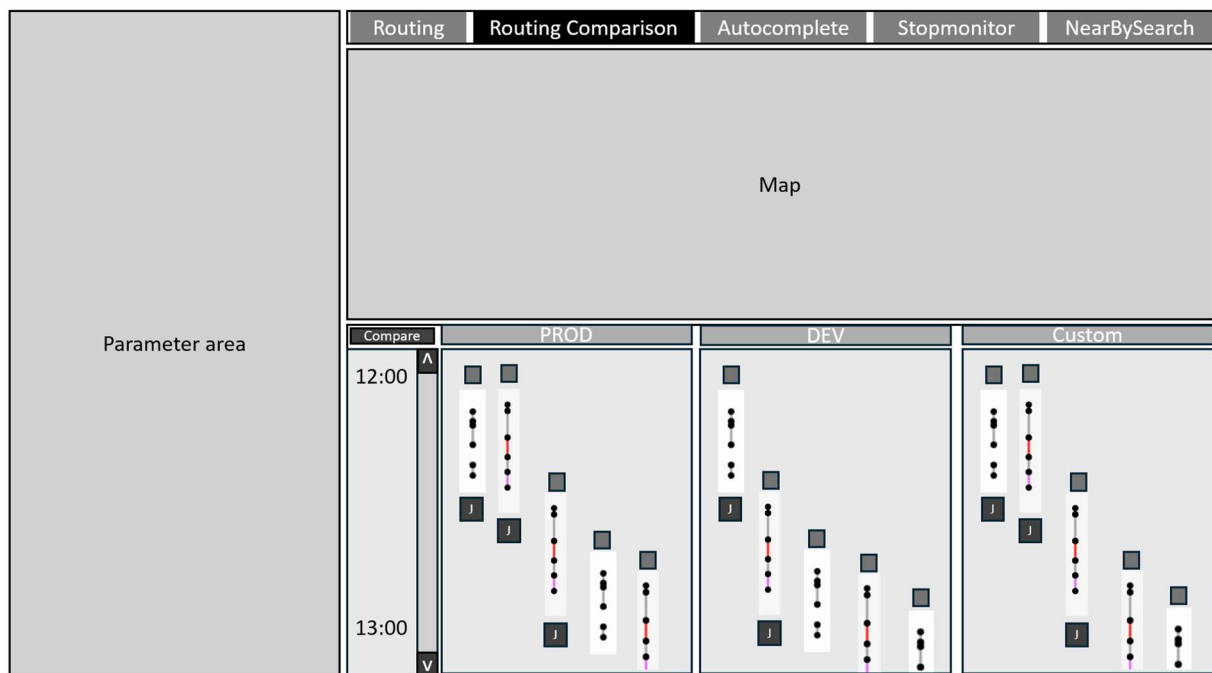


Figure 5: Routing Comparison - Side-by-side vertical

6.4 Horizontal Overview

- All itineraries shall be displayed in a single list
- A color code shall identify the source environment
- A legend shall map colors to APIs

Interaction behavior shall be consistent with other layouts.

- #0072B2 for the first API
- #E69F00 for the second API
- #009E73 for the third API

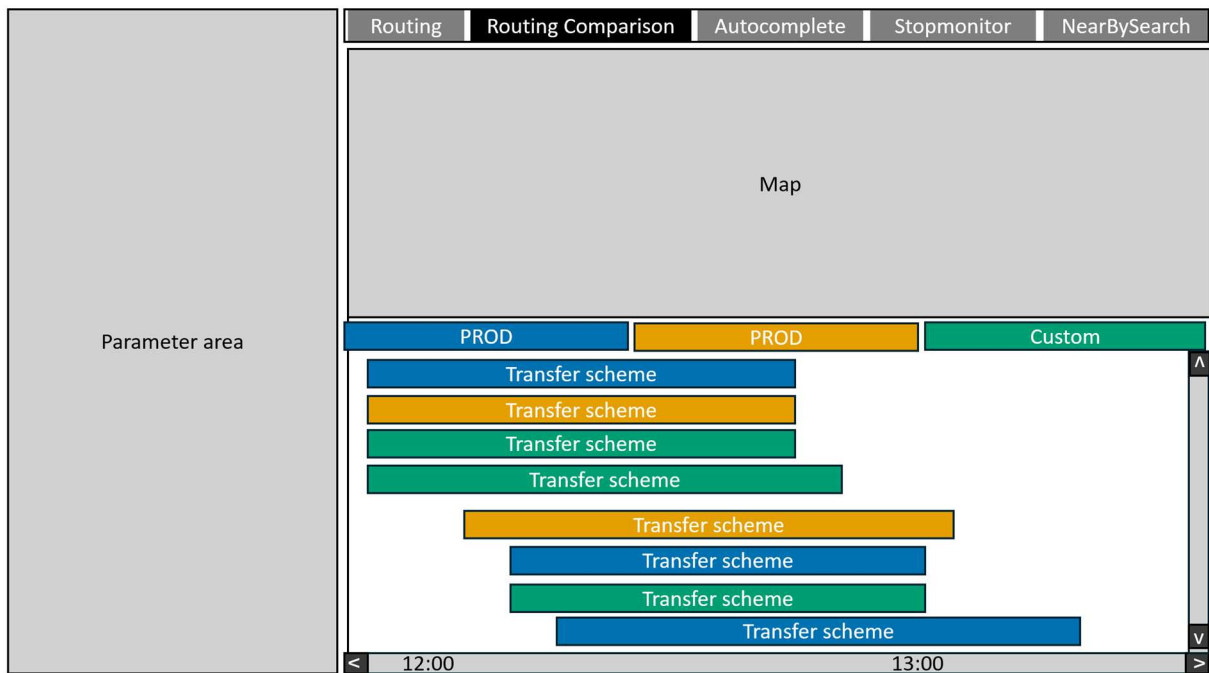


Figure 6: Routing comparison - Horizontal overview

6.5 Detail comparison view

The system shall allow up to three itineraries to be selected for comparison.

For each itinerary:

- A dedicated column shall be displayed
- A unique color shall be assigned
- Route shapes shall be displayed simultaneously on the map

The user shall be able to exit the comparison view and return to the overview.

The following colors shall be used:

- #0072B2 for the first itinerary
- #E69F00 for the second itinerary
- #009E73 for the third itinerary

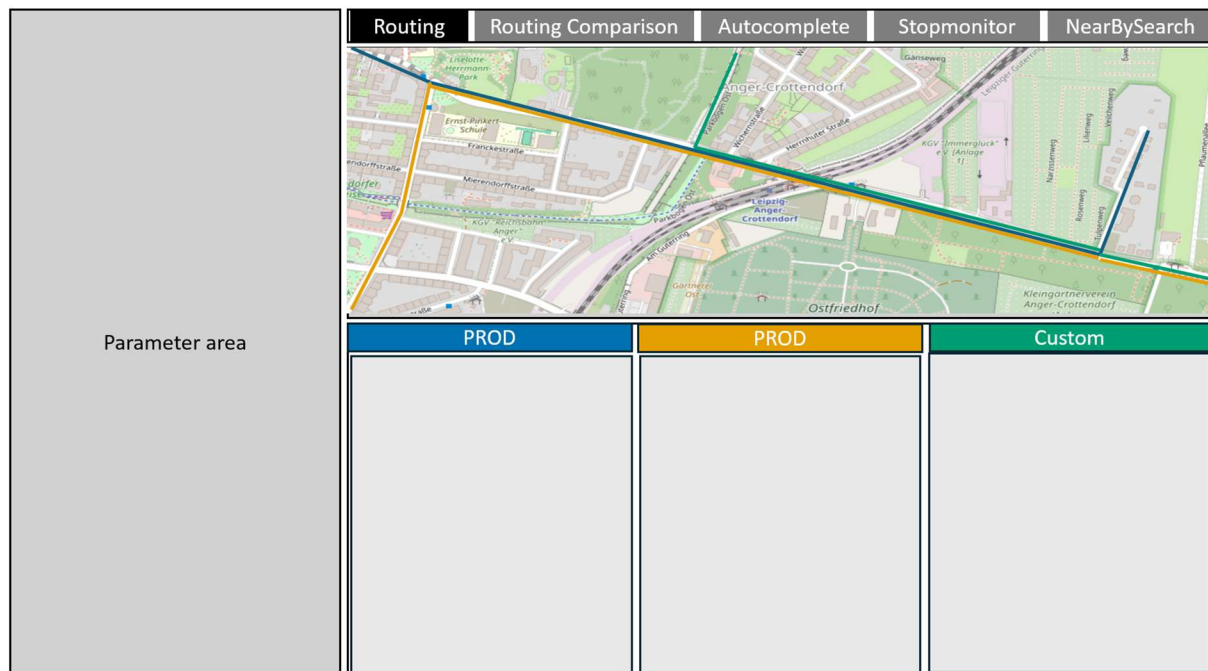


Figure 7: Routing comparison - Detail comparison view

7 Technical Context

The application shall run within the LVB technical ecosystem.

Access to required infrastructure and services shall be provided by LVB.

The application shall be implemented using Next.js.

Modules developed in a previous project and extended by a working student may be reused.